

MAINCRAFTS INTERNSHIP - TASK 1

VLSI Design Flow, Logic Gate Truth Tables and Vivado Simulation Results

1. VLSI Design Flow

Specification: Define functional, timing, power, and area requirements.

Architecture Design: Divide the system into functional hardware blocks.

RTL Design: Describe the hardware using Verilog HDL.

Functional Verification: Verify functionality using testbenches and simulation.

Logic Synthesis: Convert RTL into a gate-level netlist.

Floorplanning: Arrange major blocks on the chip layout.

Placement: Place standard cells in the allocated area.

Clock Tree Synthesis (CTS): Create a balanced clock distribution network.

Routing: Connect all components using metal interconnections.

Physical Verification: Perform DRC, LVS, and timing analysis.

Fabrication: Manufacture the integrated circuit.

Testing & Packaging: Test the fabricated IC and package it for deployment.

2. Truth Tables

AND Gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate

A	B	Y
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0	0	1
0	1	0
1	0	0
1	1	0

XOR Gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

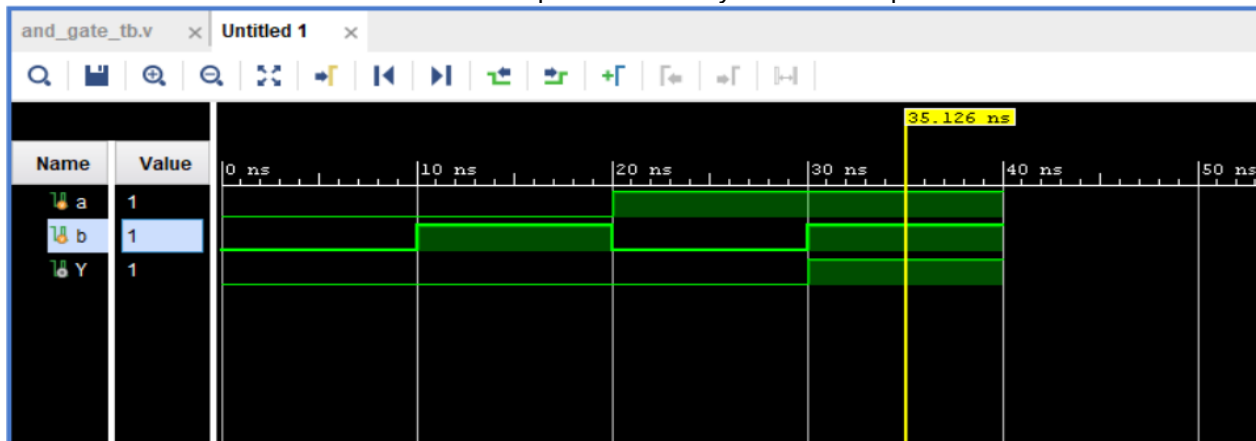
Half Adder

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

3. Vivado Simulation Results and Observations

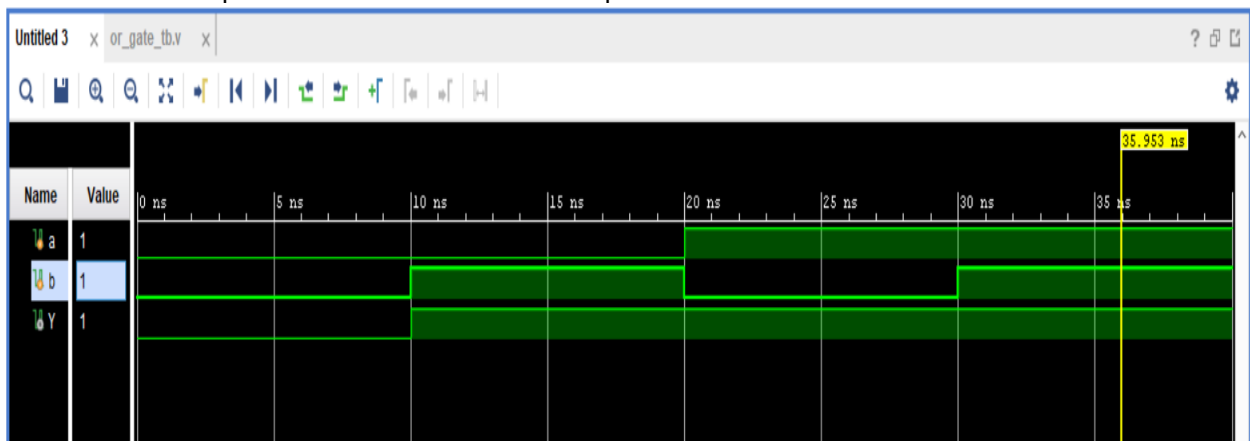
AND Gate

Observation: The simulation verifies that the output is HIGH only when both inputs are HIGH.



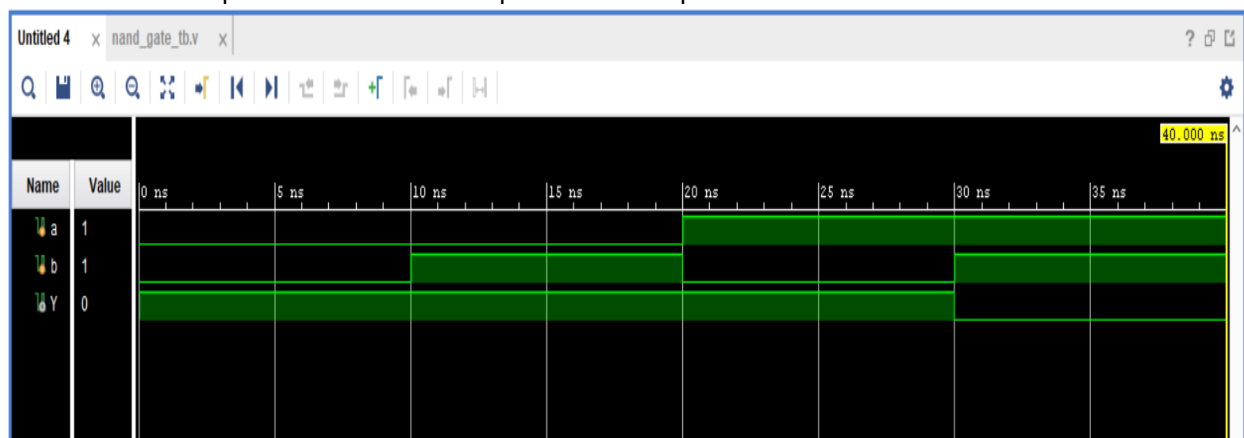
OR Gate

Observation: The output is HIGH when at least one input is HIGH.



NAND Gate

Observation: The output remains HIGH except when both inputs are HIGH.



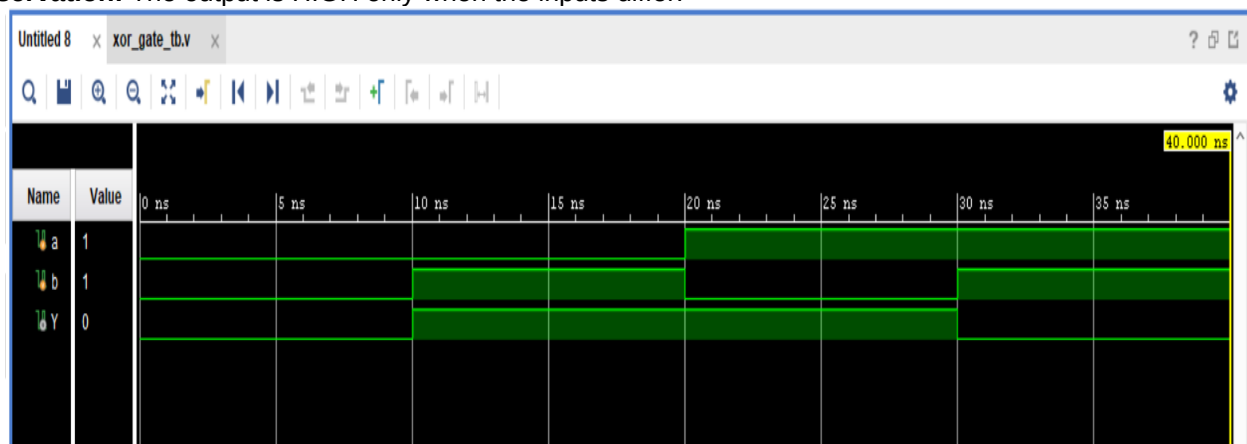
NOR Gate

Observation: The output is HIGH only when both inputs are LOW.



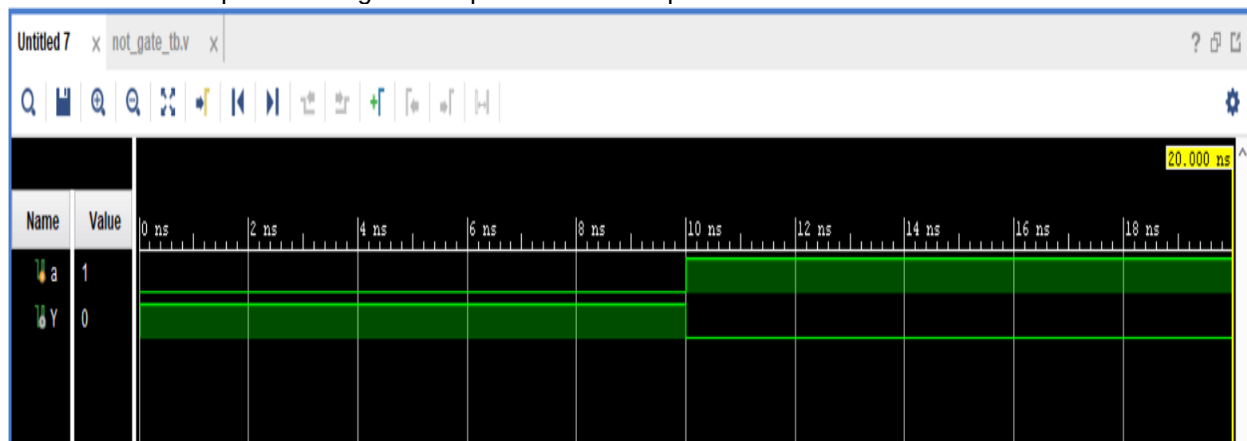
XOR Gate

Observation: The output is HIGH only when the inputs differ.



NOT Gate

Observation: The output is the logical complement of the input.



Half Adder

Observation: The simulation confirms $\text{Sum} = A \text{ XOR } B$ and $\text{Carry} = A \text{ AND } B$.

